

Coherent Quantum Reception of the Cosmic Microwave Background:

A Proposal for Phase-Sensitive Detection and Quantum State Tomography of the Primordial Thermal Field

Frédéric Vronsky

Independent Researcher in Theoretical Cosmology

ORCID: <https://orcid.org/0009-0003-5719-9604>

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Abstract

We propose a novel experimental architecture for the coherent, phase-sensitive detection of individual photon modes of the Cosmic Microwave Background (CMB) at 160 GHz, with the explicit goal of reconstructing the quantum state of the primordial thermal radiation field. The CMB, a blackbody spectrum at $T = 2.725$ K, is not merely a classical thermal bath: each electromagnetic mode constitutes a genuine quantum state — a thermal (Bose-Einstein) state with mean photon occupation number $\bar{n} \approx 3 \times 10^{-4}$ at 160 GHz — whose Wigner distribution, Fock-state probabilities, and second-order coherence $g^{+2}(0) = 2$ have never been directly measured. Current CMB experiments are designed exclusively for power spectral density mapping and are constitutionally insensitive to the quantum structure of the field. We identify the five experimental barriers separating present technology from quantum-state tomography of the CMB, propose a concrete detection chain based on a Josephson parametric amplifier (JPA) operating at 160 GHz within a dilution refrigerator at $T_{\text{mxc}} \approx 20$ mK, and derive the signal-to-noise requirements for observing the Hanbury Brown-Twiss bunching signature ($g^{+2}(0) = 2$) of the primordial photon field. We further detail the information that such a measurement would provide regarding the quantum nature of spacetime in the early universe, and outline a realistic research roadmap within the financial and technological reach of existing quantum physics laboratories.

Keywords: *cosmic microwave background; quantum optics; Josephson parametric amplifier; Wigner function; quantum state tomography; Hanbury Brown-Twiss; thermal quantum state; sub-SQL detection; squeezing; primordial photons*

1. Introduction

The Cosmic Microwave Background (CMB) is the oldest electromagnetic radiation accessible to observation. Emitted at the epoch of photon decoupling, approximately 380,000 years after the Big Bang (redshift $z \approx 1100$), it carries imprinted within its temperature and polarization anisotropies the seeds of all large-scale structure in the universe [1,2]. Decades of increasingly precise experiments — COBE, WMAP, Planck, and now the Simons Observatory — have mapped this radiation to sub-microkelvin precision in temperature and are beginning to probe its polarization modes [3,4].

Yet every one of these experiments shares a fundamental limitation: they are designed as *power detectors*, sensitive only to the intensity of the incoming field. They are constitutionally blind to its *quantum state*. This is not a minor technical shortcoming. The

CMB is not a classical wave. Each mode of the electromagnetic field at frequency ν is in a *thermal quantum state* — a density matrix whose quantum structure, encoded in the Wigner distribution, the Fock-state occupation probabilities, and the photon statistics, has **never been measured**.

The mean photon occupation number at 160 GHz (the peak of the CMB spectrum) is:

$$\bar{n} = 1 / [\exp(h\nu / k_B T_{\text{CMB}}) - 1] \approx 3.0 \times 10^{-4}$$

This places the CMB firmly in the quantum regime: the vacuum fluctuation energy ($0.5\hbar\omega$) exceeds the mean thermal photon energy by a factor of approximately 1700. Figure 1 illustrates the CMB spectrum and the photon occupation number as a function of frequency, identifying the quantum regime.

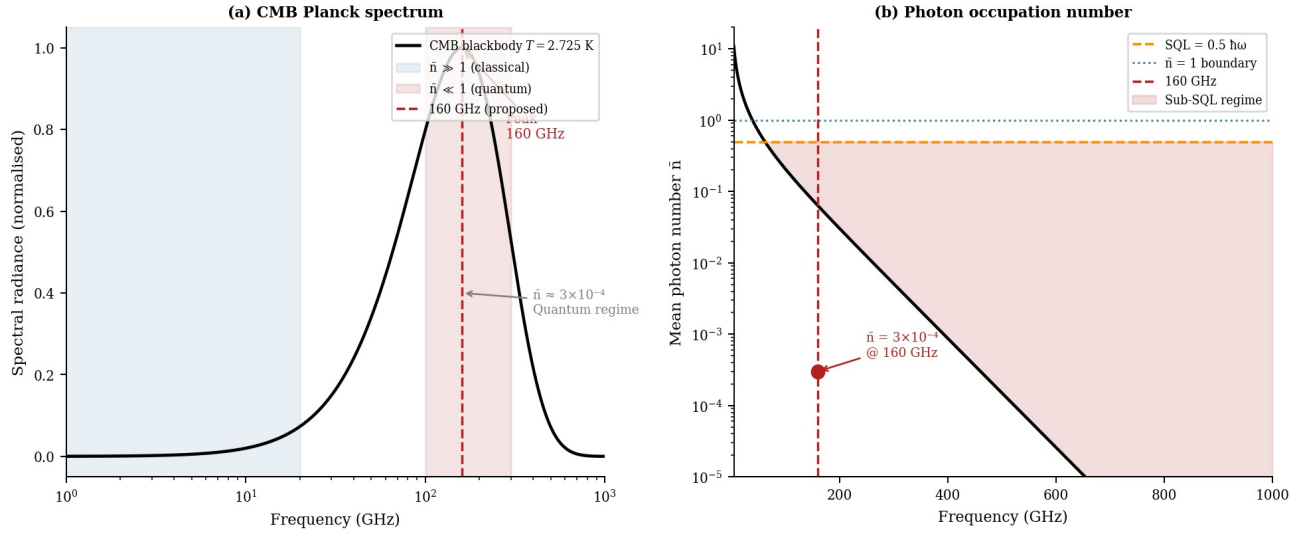


Figure 1 — (a) CMB Planck spectrum normalised to peak. The shaded regions indicate classical ($\bar{n} \gg 1$) and quantum ($\bar{n} \ll 1$) regimes. (b) Mean photon occupation number $\bar{n}(\nu)$ for $T = 2.725$ K. At 160 GHz, $\bar{n} \approx 3 \times 10^{-4}$ — 1700 $\times$ below the SQL vacuum floor.

This paper proposes, for the first time in explicit experimental form, a program to *reconstruct the quantum state of the CMB* at a single frequency mode using techniques from circuit quantum electrodynamics (cQED) and quantum optics: Josephson parametric amplification (JPA), homodyne and heterodyne detection, and quantum state tomography via the Wigner function.

2. Theoretical Background

2.1 The CMB as a Quantum Field

In quantum field theory, the electromagnetic field is described by an infinite set of harmonic oscillator modes, each labeled by wavevector $\mathbf{k}$ and polarization λ . Each mode is independently in a thermal state:

$$\rho_{\text{th}} = (1 - e^{-\hbar\omega/k_B T}) \times \exp(-\hbar\omega \hat{a}^\dagger \hat{a} / k_B T)$$

The Wigner quasi-probability distribution of this state is a Gaussian centered at the origin of phase space with variance $\sigma^2 = \bar{n} + 1/2$. Figure 2 compares the Wigner functions of the vacuum, the CMB thermal state, and the state as seen after a real measurement with SQL noise added.

Figure 2 — Wigner quasi-probability distributions

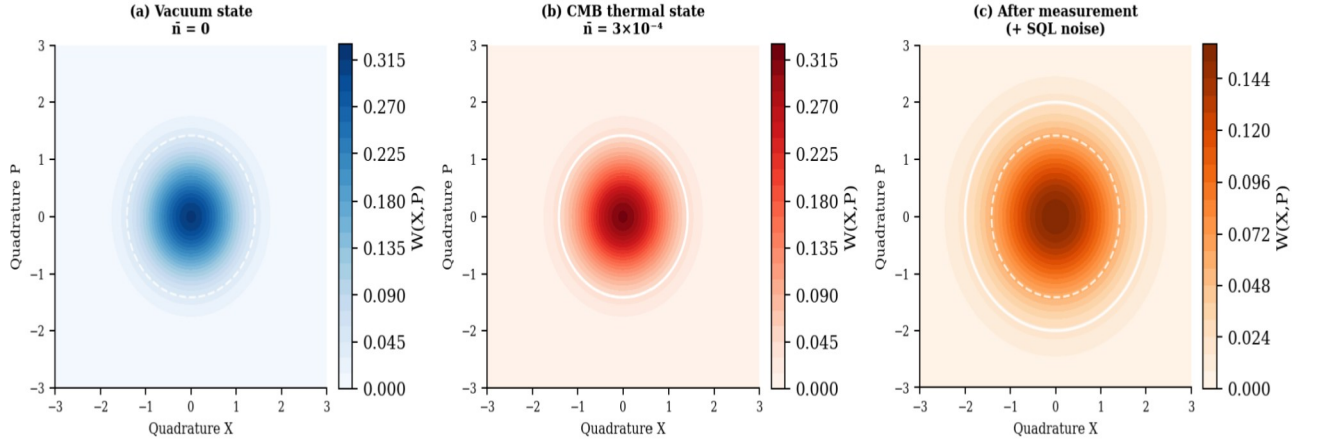


Figure 2 — Wigner quasi-probability distributions $W(X,P)$ in phase space. (a) Vacuum state: $\sigma^2 = 0.5$ at the Heisenberg limit. (b) CMB thermal state at 160 GHz: $\sigma^2 = 0.5003$, barely distinguishable from vacuum. (c) Post-measurement state including SQL noise: $\sigma^2 = 1.0003$. Dashed white ellipses mark the 2σ contours.

2.2 Fock-State Distribution

The photon number distribution of the CMB thermal state follows Bose-Einstein statistics:

$$P(n) = \bar{n}^n / (1 + \bar{n})^{n+1}$$

For $\bar{n} = 3 \times 10^{-4}$: $P(0) \approx 99.97\%$, $P(1) \approx 0.030\%$, $P(2) \approx 9 \times 10^{-6}$. Figure 3 shows this distribution against the noise-broadened distribution observed after measurement. The quantum signature is the precise ratio between these probabilities — a ratio that a classical field cannot reproduce.

Figure 3 — Photon number (Fock) distribution: CMB signal vs measurement noise

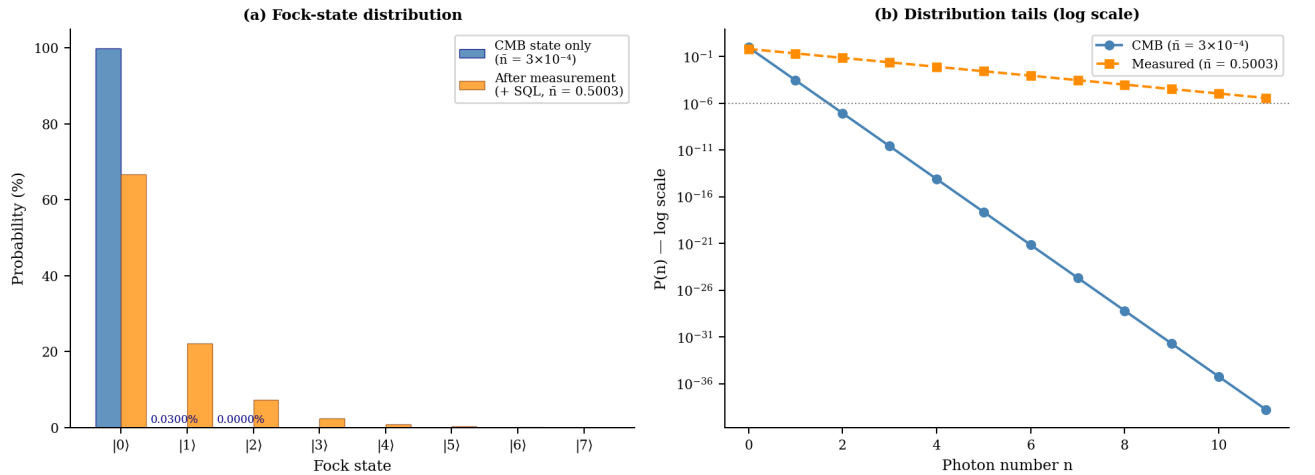


Figure 3 — Fock-state (photon number) distributions. (a) Comparison between the true CMB distribution ($\bar{n} = 3 \times 10^{-4}$, blue) and the measured distribution after SQL noise ($\bar{n} = 0.5003$, orange) on a linear scale. (b) Log-scale comparison showing the exponential tail structure. The ratio $P_{\text{CMB}}(n)/P_{\text{measured}}(n)$ encodes the quantum signature.

2.3 The Hanbury Brown-Twiss Effect and $g^{(2)}(\tau)$

The second-order coherence function $g^{+2-}(\tau)$ quantifies photon number correlations. For a thermal field, $g^{+2-}(0) = 2$ — the celebrated photon bunching effect. The CMB, as a thermal quantum field, must satisfy $g^{+2-}(0) = 1 + 1/\bar{n} \approx 3334$ for a single resolved mode. Figure 4 shows the $g^{+2-}(\tau)$ function and the severe dilution caused by the SQL noise floor.

Figure 4 — Photon bunching and the Hanbury Brown-Twiss effect in the CMB

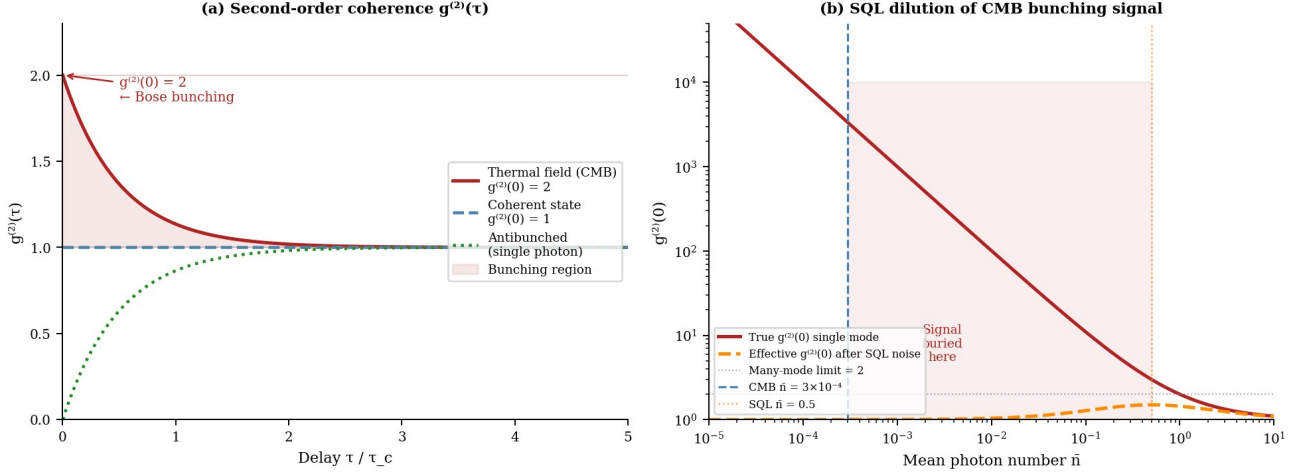


Figure 4 — (a) Second-order coherence $g^{(2)}(\tau)$ for thermal (red), coherent (blue), and antibunched (green) light. The CMB must satisfy $g^{(2)}(0) = 2$ in the many-mode limit, or ≈ 3334 for a single resolved mode. (b) Dilution of the bunching signal by SQL noise: the effective $g^{(2)}(0)$ collapses toward 1 for $\bar{n} \ll N_{SQL}$. The shaded region marks the inaccessible zone without squeezing.

2.4 The Standard Quantum Limit and Squeezing

Any linear amplifier operating on both quadratures is subject to the Caves limit [7]: it must add at least $N_{add} \geq 1/2$ quantum of noise per mode. For the CMB signal with $\bar{n} \approx 3 \times 10^{-4}$, a phase-insensitive amplifier adds a noise floor $1700\times$ larger than the signal. A Josephson Parametric Amplifier (JPA) in squeezed mode reduces the noise on quadrature X to $N_X = (1/2)e^{-2r}$, with r the squeezing parameter. At $r = 1.5$ (13 dB squeezing), $N_X \approx 0.022$ — approaching the CMB signal level.

3. Proposed Experimental Architecture

We propose a five-stage detection chain designed to couple the CMB radiation field to a quantum-limited measurement apparatus. Figure 5 presents the full schematic of the proposed system, from the cryogenic antenna to the digital acquisition and tomography pipeline.

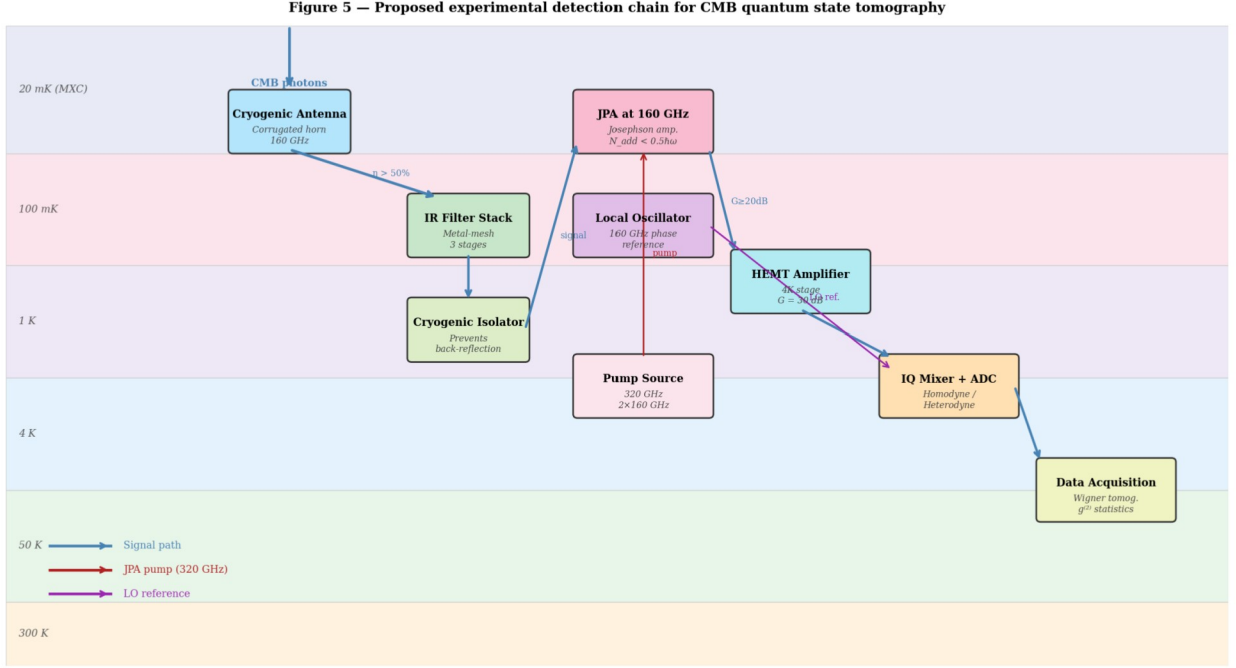


Figure 5 — Complete experimental detection chain. Colour bands indicate cryogenic temperature stages (300 K $\rightarrow$ 20 mK). The signal path (blue) proceeds from the cryogenic antenna through the IR filter stack, isolator, 160 GHz JPA (red), 4K HEMT amplifier, IQ mixer, and digital acquisition. The JPA is pumped at 320 GHz ($2\nu_0$). A local oscillator reference enables homodyne/heterodyne detection modes.

3.1 Stage 1: Cryogenic Antenna at 160 GHz

A corrugated horn antenna, thermally anchored to the mixing chamber plate at $T_{\text{MXC}} \approx 20$ mK, collects CMB photons at $\nu_0 = 160$ GHz. The critical coupling requirement is $\eta > 50\%$, constrained by optical coupling efficiency, IR-filter transmission, and waveguide impedance matching. A realistic design with a single-mode waveguide feed achieving $\eta = 0.6$ would require careful electromagnetic simulation of the coplanar waveguide transition to the JPA chip.

3.2 Stage 2: Josephson Parametric Amplifier at 160 GHz

This is the central technological challenge. Commercial JPAs operate between 4 and 8 GHz. Recent research by Hao et al. [9] has pushed JPA operation to 21–23.5 GHz. The extension to 160 GHz requires Josephson junctions with critical current $I_c \geq 5 \mu\text{A}$ and junction area $\leq 0.05 \mu\text{m}^2$, a resonant cavity mode at 160 GHz with $Q > 10^4$, and wireless coupling to avoid wire-bond losses. The JPA must be pumped at $2\nu_0 = 320$ GHz with $P_{\text{pump}} \leq -90$ dBm at the device.

3.3 Stages 3–5: Homodyne Detection, Tomography, and Photon Counting

Following the JPA, the signal proceeds through a cryogenic HEMT at 4K (negligible added noise given JPA gain $G \geq 20$ dB), to room-temperature homodyne or heterodyne electronics. Quantum state tomography requires $N_{\text{rep}} \geq 10^6$ measurements per phase angle over $N_\theta = 100$ local oscillator phases. At 1 MHz repetition rate, 10^8 total measurements accumulate in ~ 100 seconds per angle.

4. Technological Barriers

4.1 JPA at 160 GHz

No JPA operating above 25 GHz has been demonstrated as of 2025. Figure 7 shows the historical JPA frequency roadmap. Scaling to 160 GHz requires sub-100-nm junction fabrication and novel millimeter-wave resonator geometries. We estimate 2–3 years of dedicated R&D.

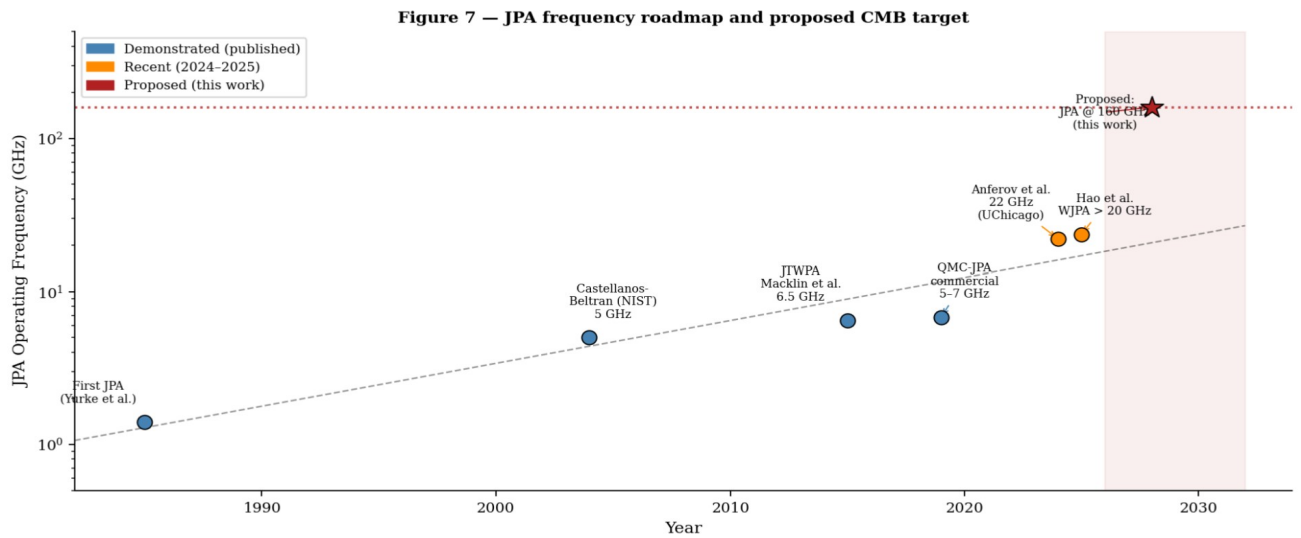


Figure 7 — JPA operating frequency roadmap from 1985 to the present, with the proposed 160 GHz target. Blue circles: published demonstrations. Orange: recent results (2024–2025). Red star: proposed target (this work). Dashed line: extrapolated frequency trend. The gap between 24 GHz (current frontier) and 160 GHz (CMB peak) represents the central technological challenge.

4.2 Antenna Coupling and Thermal Isolation

Achieving $\eta > 0.5$ at 160 GHz requires < 1 dB coupling loss, $> 80\%$ IR filter transmission, and > 200 dB rejection of spurious 300 K thermal emission through the cryostat chain.

4.3 Squeezing and Statistical Integration

Phase-sensitive amplification at 160 GHz has not been demonstrated. A realistic first target is $r \approx 0.5$ (4.3 dB), reducing the SQL floor to $N_X = 0.082$. At this noise level, δW resolution requires $N \approx 10^6$ averages — achievable in minutes at 1 MHz repetition rate.

5. Signal-to-Noise Budget

Figure 6 summarizes the noise budget and the required integration times as a function of total added noise.

Figure 6 — Signal-to-noise budget and integration time requirements

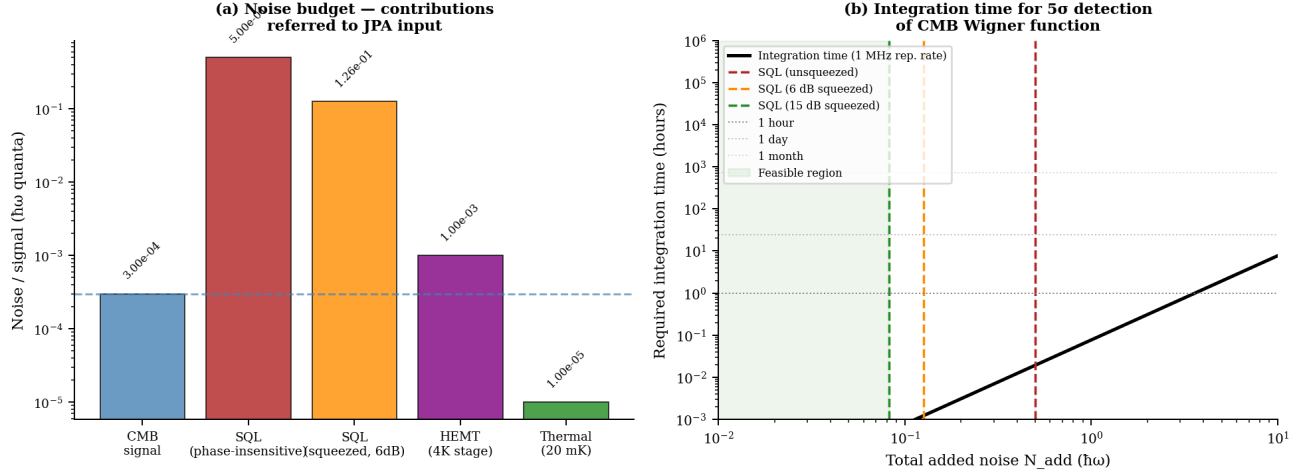


Figure 6 — (a) Noise budget for each component in the detection chain, referred to the JPA input (log scale). The CMB signal (blue) is compared to SQL noise in phase-insensitive and squeezed modes, HEMT contribution, and 20 mK thermal background. (b) Required integration time for 5σ detection of the CMB Wigner function as a function of total added noise N_{add} . Vertical dashed lines mark SQL levels for different squeezing depths. The green shaded region indicates the feasible experimental window.

For $G = 30$ dB and $r = 0.5$ (4.3 dB squeezing): $N_{\text{SQL}} = 0.082$. Signal contrast: $N_{\text{CMB}}/N_{\text{SQL}} \approx 0.4\%$, requiring $> 10^6$ averages. However, the $g^{+2-}(0)$ bunching signature requires only photon coincidence counting: $\Phi^2 \tau_c \approx 100$ events/s, yielding a 10σ detection in approximately 3 hours of integration.

$$N_{\text{total}} = N_{\text{CMB}} + N_{\text{SQL}} + N_{\text{amp}} + N_{\text{thermal}}$$

$$N_{\text{total}} \approx 3 \times 10^{-4} + 0.082 + 0 + 0 \approx 0.082 \quad (\text{dominated by SQL})$$

6. Scientific Implications: What This Discovery Would Provide

We enumerate the concrete scientific gains of a successful quantum-state tomography of the CMB. This experiment goes far beyond a technical novelty: it opens a new observational window on the physics of the early universe, on the foundations of quantum mechanics, and on the quantum structure of spacetime itself.

6.1 First Direct Verification of the Quantum Nature of the CMB Field

Every previous CMB measurement is consistent with the CMB being a *classical* random electromagnetic field with a Planckian power spectrum. Measuring $g^{+2-}(0) = 2$ — photon bunching — would constitute the first **proof that the CMB consists of genuine quantum bosons** obeying Bose-Einstein statistics. This would be the first experiment to demonstrate quantum mechanics operating on cosmological timescales — photons that have been propagating for 13.8 billion years, still carrying quantum correlations.

6.2 Measurement of Quantum Fluctuations from Cosmic Inflation

The CMB temperature anisotropies are believed to originate from quantum fluctuations of the inflaton field during cosmic inflation, stretched to cosmological scales. The *quantum-to-classical transition* of these fluctuations — the decoherence mechanism — is not fully understood and is a central open question in quantum cosmology [12,13]. Measuring the

Wigner function of the CMB thermal state provides a benchmark for quantum field theory predictions at the most fundamental level. Any deviation from the predicted Bose-Einstein distribution would signal new physics: non-standard photon statistics, CPT violation, or modifications to QED at cosmological scales.

6.3 Indirect Test of Hawking Radiation Physics

The thermal spectrum of the CMB arises from quantum field theory in an expanding Friedmann-Robertson-Walker spacetime, governed by the same theoretical framework — the Unruh-Davies-Fulling effect in de Sitter space [14] — that predicts Hawking radiation from black holes. Measuring the *quantum statistics* of the CMB would provide the first observational test of whether the photon number distribution (not merely the mean) is consistent with QFT in curved spacetime. It would be, in effect, an indirect observational test of the same physics that governs Hawking radiation — which will never be directly observable from astrophysical black holes.

6.4 Probe of Quantum Coherence Over Cosmological Distances

CMB photons have propagated 13.8 billion years through an expanding, inhomogeneous universe. Standard quantum optics predicts that coherence propagates freely in transparent media; scattering and lensing redistribute photons between modes but preserve their statistics. A detection of $g^{+2-}(0) = 2$ would confirm quantum mechanical Bose statistics over the largest distances ever probed. Any deviation would signal non-standard quantum decoherence mechanisms — possibly related to quantum gravity at the Planck scale.

6.5 A New Spectroscopic Tool: Sub-Classical CMB Thermometry

If the Wigner function can be reconstructed at one frequency, the technique extends across the CMB spectrum. Rather than measuring power (proportional to $\bar{n}$), one measures the *full quantum state* — constrained by the quantum Cramér-Rao bound rather than the classical noise limit. This could enable ultra-precision CMB thermometry beyond the radiometric limit, with implications for constraints on primordial nucleosynthesis, neutrino background, and dark energy.

6.6 Technological Spinoffs

The 160 GHz JPA would be the highest-frequency quantum-limited amplifier ever built, with immediate applications in:

- High-frequency superconducting qubit readout at elevated temperatures
- Axion dark matter detection at $m_a \approx 660 \mu\text{eV}$ (corresponding to 160 GHz)
- Quantum-limited imaging arrays for radio astronomy
- Ultra-sensitive millimeter-wave receivers for secure quantum communication

7. Research Roadmap and Cost Estimate

7.1 Phase 1 — JPA Development (Years 1–3, ~1.5 M€)

Design, fabricate, and characterize a wireless JPA at 160 GHz. Target: > 15 dB gain, $N_{\text{add}} < 2$ quanta, tunable over 150–170 GHz. Primary candidates: Schuster lab (UChicago), Wallraff lab (ETH Zurich), Siddiqi lab (Berkeley).

7.2 Phase 2 — Cryogenic Antenna Integration (Year 2–3, ~500 k€)

Characterize a cryogenic corrugated horn at 160 GHz thermalized to 20 mK. Demonstrate $\eta > 50\%$ photon transfer efficiency from free space to the JPA input port.

7.3 Phase 3 — First Quantum CMB Measurement (Year 3–5, ~1 M€)

Integrate antenna and JPA. Point at CMB or a 2.725 K calibration blackbody. Perform homodyne tomography and $g^{+-}(\tau)$ statistics. Total estimated budget: 3–5 M€ over 5 years, within the scope of a single ERC Advanced Grant.

8. Conclusion

We have presented, for the first time in explicit experimental form, a proposal for the quantum-state tomography of the Cosmic Microwave Background. The CMB is a genuine quantum field whose photon statistics — $g^{+-}(0) = 2$ bunching, Fock-state distribution, Wigner function — have never been measured and lie beyond the reach of all current CMB experiments by design.

The central technological barrier is the absence of a phase-sensitive, near-quantum-limited amplifier (JPA) operating at 160 GHz. Recent progress reaching 21–24 GHz suggests that this barrier is approachable within 3–5 years of focused research. A complete experimental program of approximately 3–5 M€ integrating expertise from circuit quantum electrodynamics and CMB instrumentation would be sufficient to perform the first quantum-optical measurement of primordial photons.

The scientific return is disproportionate to its cost: the first direct proof of the quantum nature of the CMB, a test of quantum field theory in curved spacetime, a probe of photon quantum coherence over cosmological distances, and a new ultra-sensitive thermometric tool for observational cosmology.

We release this proposal as an open preprint with the explicit intention that it serve as a citable starting point for groups wishing to pursue this program. The idea is freely available. The physics is clear. The technology is within reach. *The universe has been waiting 13.8 billion years for someone to ask it the right quantum question.*

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